

MUHTASIM AL AHSAN

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Ghatarchar, Keraniganj, Dhaka, Bangladesh

Research Profile

Undergraduate Student in Computer Science & Engineering from State University of Bangladesh, CGPA 3.99/4.00, working in the field of intersection between AI, Robotics and Biomedical Engineering. Conference published author in deep learning; experience with prototype development of biomedical instrumentation, such as PCR thermal cyclers, cell culture incubators, EMG controlled robotic arm interfaces. Highly skilled in programming (Python, C++), highly interested in research training in AI-driven biomedical and robotic systems with research intensive supervision.

Education

- **Bachelor of Science (Hons.) in Computer Science & Engineering**, State University of Bangladesh, Dhaka – CGPA: 3.99/4.00 (*11th Semester*) | September 2022 – Present
- **Higher Secondary Certificate (HSC)** – Science, Dinajpur Govt. College – GPA: 5.00/5.00 | January 2021
- **Secondary School Certificate (SSC)** – Science, Dinajpur Zila School – GPA: 5.00/5.00 | March 2018

Research Projects & Publications

- **Conference Paper:** Al Ahsan, M., et al. (2025). “*A Deep Space Flexible Blind Convolutional Neural Network for Super Compressed JPEG Artifacts Removal*,” 2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), Rangpur, Bangladesh, pp. 1–6. **doi: 10.1109/QPAIN66474.2025.11171870.**
- **Undergraduate Project:** *Design and Implementation of an Intelligent sEMG-Based Human–Machine Interface for Multi-DOF Robotic Arm Control* – developed an electromyography-driven control system enabling multi-degree-of-freedom actuation of a robotic arm through surface EMG signal processing.

Professional Experience

- **Biomedical Engineering Assistant**, Rufaida BioMeds, Aftabnagar, Dhaka (March 2025 – April 2026)
 - Designed and prepared a mechanical rocker for Western Blotting applications.
 - Designed a cell culture incubator for mammalian cell culture protocols.
 - Developed a working prototype of a thermal cycler (PCR machine) for nucleic acid amplification.
- **IT Intern (Part-Time)**, Rufaida BioMeds, Aftabnagar, Dhaka (March 2024 – February 2025)
 - Upgraded and maintained the Rufaida BioMeds organisational website (rufaidabiomed.org).
 - Developed and maintained the Callovia website (callovia.com).
 - Developed and managed the CareerLink Global Education website.

Technical Skills

- **Programming Languages:** Python, C++, Java, HTML & CSS
- **Artificial Intelligence:** Deep Learning, Convolutional Neural Networks, Computer Vision
- **Computer-Aided Design & Simulation:** SolidWorks, Unreal Engine
- **Hardware & Systems:** Electronics and Mechatronics, Embedded Systems, Biomedical Instrumentation
- **Data & Tools:** Database Management Systems

Achievements & Honors

- **Honorable Mention**, 2025 ICPC Asia Dhaka Regional Contest
- **Finalist**, 2024 ICPC Asia Dhaka Regional Contest
- **Finalist**, 1st Astronomy Olympiad of Bangladesh (2019)

Leadership & Co-curricular Engagement

- **Founder & President**, *SUBRAS – State University of Bangladesh Robotics and Automation Society*
- **General Secretary**, *Computing Club of State University of Bangladesh*

Languages

- **Bengali** – Native language
- **English** – Fluent listener and speaker; proficient in reading and writing

References

- **Dr. Pintu Chandra Shill**

Professor and Chairman, *Department of Computer Science and Engineering, State University of Bangladesh*

Email: pintu.cse@sub.edu.bd

- **Dr. Md. Aminul Haque**

Associate Professor, *School of Pharmacy, BRAC University*

Email: aminul.haque@bracu.ac.bd